

Q1 - 24 January - Shift 1

The relation $R = \{(a, b) : \gcd(a, b) = 1, 2a \neq b, a, b \in \mathbb{Z}\}$

is: _____

- (1) transitive but not reflexive
- (2) symmetric but not transitive
- (3) reflexive but not symmetric
- (4) neither symmetric nor transitive

Space for your notes:

Q2 - 24 January - Shift 2

The minimum number of elements that must be added to the relation $R = \{(a, b), (b, c), (b, d)\}$ on the set $\{a, b, c, d\}$ so that it is an equivalence relation, is _____.

Space for your notes:

Q3 - 29 January - Shift 2

Let R be a relation defined on $\mathbb{N}$ as $a R b$ is $2a + 3b$ is a multiple of 5, $a, b \in \mathbb{N}$. Then R is

- (1) not reflexive
- (2) transitive but not symmetric
- (3) symmetric but not transitive
- (4) an equivalence relation

Space for your notes:

Q4 - 30 January - Shift 1

The minimum number of elements that must be added to the relation $R = \{(a, b), (b, c)\}$ on the set

$\{a, b, c\}$ so that it becomes symmetric and transitive is:

- (1) 4
- (2) 7
- (3) 5
- (4) 3

Space for your notes:

Q5 - 31 January - Shift 1

Let R be a relation on $N \times N$ defined by $(a, b) R (c, d)$ if and only if $ad(b - c) = bc(a - d)$. Then R is

- (1) symmetric but neither reflexive nor transitive
- (2) transitive but neither reflexive nor symmetric
- (3) reflexive and symmetric but not transitive
- (4) symmetric and transitive but not reflexive

Space for your notes:

Q6 - 31 January - Shift 2

Among the relations

$$S = \left\{ (a, b) : a, b \in \mathbb{R} - \{0\}, 2 + \frac{a}{b} > 0 \right\}$$

And $T = \{ (a, b) : a, b \in \mathbb{R}, a^2 - b^2 \in \mathbb{Z} \}$,

- (1) S is transitive but T is not
- (2) T is symmetric but S is not
- (3) Neither S nor T is transitive
- (4) Both S and T are symmetric

Space for your notes:

Q7 - 01 February - Shift 1

Let R be a relation on $\mathbb{R}$, given by

$R = \{ (a, b) : 3a - 3b + \sqrt{7} \text{ is an irrational number} \}$. Then R is

- (1) Reflexive but neither symmetric nor transitive
- (2) Reflexive and transitive but not symmetric
- (3) Reflexive and symmetric but not transitive
- (4) An equivalence relation

Space for your notes:

Q8 - 01 February - Shift 2

Let $P(S)$ denote the power set of $S = \{1, 2, 3, \dots, 10\}$.

Define the relations R_1 and R_2 on $P(S)$ as AR_1B if

$(A \cap B^c) \cup (B \cap A^c) = \emptyset$ and AR_2B if $A \cup B^c =$

$B \cup A^c, \forall A, B \in P(S)$. Then :

- (1) both R_1 and R_2 are equivalence relations
- (2) only R_1 is an equivalence relation
- (3) only R_2 is an equivalence relation
- (4) both R_1 and R_2 are not equivalence relations

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (4)

Q2 (13)

Q3 (4)

Q4 (2)

Q5 (1)

Q6 (2)

Q7 (1)

Q8 (1)

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Q1 (4)

Reflexive : $(a, a) \Rightarrow \gcd \text{ of } (a, a) = 1$

Which is not true for every $a \in \mathbb{Z}$.

Symmetric:

Take $a = 2, b = 1 \Rightarrow \gcd(2, 1) = 1$

Also $2a = 4 \neq b$

Now when $a = 1, b = 2 \Rightarrow \gcd(1, 2) = 1$

Also now $2a = 2 = b$

Hence $a = 2b$

$\Rightarrow R$ is not Symmetric

Transitive:

Let $a = 14, b = 19, c = 21$

$\gcd(a, b) = 1$

$\gcd(b, c) = 1$

$\gcd(a, c) = 7$

Hence not transitive

$\Rightarrow R$ is neither symmetric nor transitive.

Q2 (13)

Given $R = \{(a, b), (b, c), (b, d)\}$

In order to make it equivalence relation as per given set, R must be

$\{(a, a), (b, b), (c, c), (d, d), (a, b), (b, a), (b, c), (c, b), (b, d), (d, b), (a, c), (a, d), (c, d), (d, c), (c, a), (d, a)\}$

There already given so 13 more to be added.

Q3 (4)

$a R b \Rightarrow 5a$ is multiple of 5

So reflexive

$$a R b \Rightarrow 2a + 3b = 5\alpha,$$

Now $b R a$

$$2b + 3a = 2b + \left(\frac{5\alpha - 3b}{2} \right) \cdot 3$$

$$= \frac{15}{2}\alpha - \frac{5}{2}b = \frac{5}{2}(3\alpha - b)$$

$$= \frac{5}{2}(2a + 2b - 2\alpha)$$
$$= 5(a + b - \alpha)$$

Hence symmetric

$$a R b \Rightarrow 2a + 3b = 5\alpha.$$

$$b R c \Rightarrow 2b + 3c = 5\beta$$

$$\text{Now } 2a + 5b + 3c = 5(\alpha + \beta)$$

$$\Rightarrow 2a + 5b + 3c = 5(\alpha + \beta)$$

$$\Rightarrow 2a + 3c = 5(\alpha + \beta - b)$$

$$\Rightarrow a R c$$

Hence relation is equivalence relation.

Q4 (2)

For Symmetric $(a,b), (b,c) \in R$

$$\Rightarrow (b,a), (c,b) \in R$$

For Transitive $(a,b), (b,c) \in R$

$$\Rightarrow (a,c) \in R$$

Now

1. Symmetric

$$\therefore (a,c) \in R \Rightarrow (c,a) \in R$$

2. Transitive

$$\therefore (a,b), (b,a) \in R$$

$$\Rightarrow (a,a) \in R \text{ \& } (b,c), (c,b) \in R$$

$$\Rightarrow (b,b) \text{ \& } (c,c) \in R$$

$\therefore$ Elements to be added

$$\left\{ (b,a), (c,b), (a,c), (c,a), (a,a), (b,b), (c,c) \right\}$$

Number of elements to be added = 7

Q5 (1)

$$(a, b) R (c, d) \Rightarrow ad(b - c) = bc(a - d)$$

Symmetric:

$$(c, d) R (a, b) \Rightarrow cb(d - a) = da(c - b) \Rightarrow$$

Symmetric

Reflexive:

$$(a, b) R (a, b) \Rightarrow ab(b - a) \neq ba(a - b) \Rightarrow$$

Not reflexive

Transitive: $(2, 3) R (3, 2)$ and $(3, 2) R (5, 30)$ but

$$((2, 3), (5, 30)) \notin R \Rightarrow \text{Not transitive}$$

Q6 (2)

For relation $T = a^2 - b^2 = -I$

Then, (b, a) on relation R

$$\Rightarrow b^2 - a^2 = -I$$

$\therefore T$ is symmetric

$$S = \left\{ (a, b) : a, b \in \mathbb{R} - \{0\}, 2 + \frac{a}{b} > 0 \right\}$$

$$2 + \frac{a}{b} > 0 \Rightarrow \frac{a}{b} > -2, \Rightarrow \frac{b}{a} < \frac{-1}{2}$$

If $(b, a) \in S$ then

$$2 + \frac{b}{a} \text{ not necessarily positive}$$

$\therefore S$ is not symmetric

Q7 (1)

Check for reflexivity:

As $3(a - a) + \sqrt{7} = \sqrt{7}$ which belongs to relation
so relation is reflexive

Check for symmetric:

Take $a = \frac{\sqrt{7}}{3}, b = 0$

Now $(a, b) \in R$ but $(b, a) \notin R$

As $3(b - a) + \sqrt{7} = 0$ which is rational so relation
is not symmetric.

Check for Transitivity:

Take (a, b) as $\left(\frac{\sqrt{7}}{3}, 1\right)$

& (b, c) as $\left(1, \frac{2\sqrt{7}}{3}\right)$

So now $(a, b) \in R$ & $(b, c) \in R$ but $(a, c) \notin R$
which means relation is not transitive

Q8 (1)

$$S = \{1, 2, 3, \dots, 10\}$$

$P(S)$ = power set of S

$$A, B \Rightarrow (A \cap \bar{B}) \cup (\bar{A} \cap B) = \phi$$

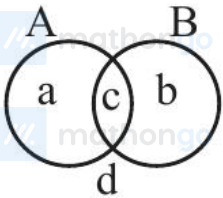
R_1 is reflexive, symmetric

For transitive

$$(A \cap \bar{B}) \cup (\bar{A} \cap B) = \phi ; \{a\} = \phi = \{b\} \quad A = B$$

$$(B \cap \bar{C}) \cup (\bar{B} \cap C) = \phi \therefore B = C$$

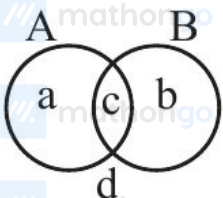
$\therefore A = C$ equivalence.



$$R_2 \equiv A \cup \bar{B} = \bar{A} \cup B$$

$R_2 \rightarrow$ Reflexive, symmetric

for transitive



$$A \cup \bar{B} = \bar{A} \cup B \Rightarrow \{a, c, d\} = \{b, c, d\}$$

$$\{a\} = \{b\} \therefore A = B$$

$$B \cup \bar{C} = \bar{B} \cup C \Rightarrow B = C$$

$$\therefore A = C \quad \therefore A \cup \bar{C} = \bar{A} \cup C \therefore \text{Equivalence}$$